



Function manual

Ice bath Heat Pump Controller
with Wi-Fi Function
(Model No. CCIB001-CC1177D)

Thank you for selecting our air to water heat pump!

This manual is for Ice bath heat pump.

The appearance of the heat pump you buy from us maybe is different from on the manual, but it doesn't affect operating and using.

Before using heat pump, please read the manual carefully.

Please safe keeping the manual for checking in future!

Simple controller of Ice bath heat pump heater Manual

1 General description

- 1.1 Three operation modes: automation, cooling and heating.
- 1.2 Single compressor system.
- 1.3 Controlled by a wire control panel.
- 1.4 System operation parameters and set parameters can be displayed and changed.
- 1.5 With automatic protection and automatic failure alarm function;
- 1.6 System protection: 3 minute compressor protection; high/low pressure protection; sensor protection; water flow detection, etc.
- 1.7 Communication distance between the unit and the wire control is not shorter than 100m;
- 1.8 With strong anti-interference, stable and reliable performance.
- 1.9 When a wire control panel is connected and ON/OFF switch closes. Unit's ON/OFF will be controlled by panel.
- 1.10 While there is no wire control panel connected, Unit will be ON if ON/OFF switch is closed and unit will be OFF if ON/OFF switch is opened.
- 1.11 Electric parameter:
 - Power supply: $220 \pm 10\%$ (VAC), 50/60Hz
 - Senor type: NTC($5K\Omega/25^{\circ}C$, B value 3470K)
 - Mounting size: 99×116×30mm
 - Display size: 86×86×18mm
 - Measuring range: $-9^{\circ}C \sim 99^{\circ}C$
 - Controlling range: $-9^{\circ}C \sim 99^{\circ}C$
 - Port output capacity: Compressor: 30A/250VAC
 - Others: 5A/250VAC



2 Control panel manual

Interface

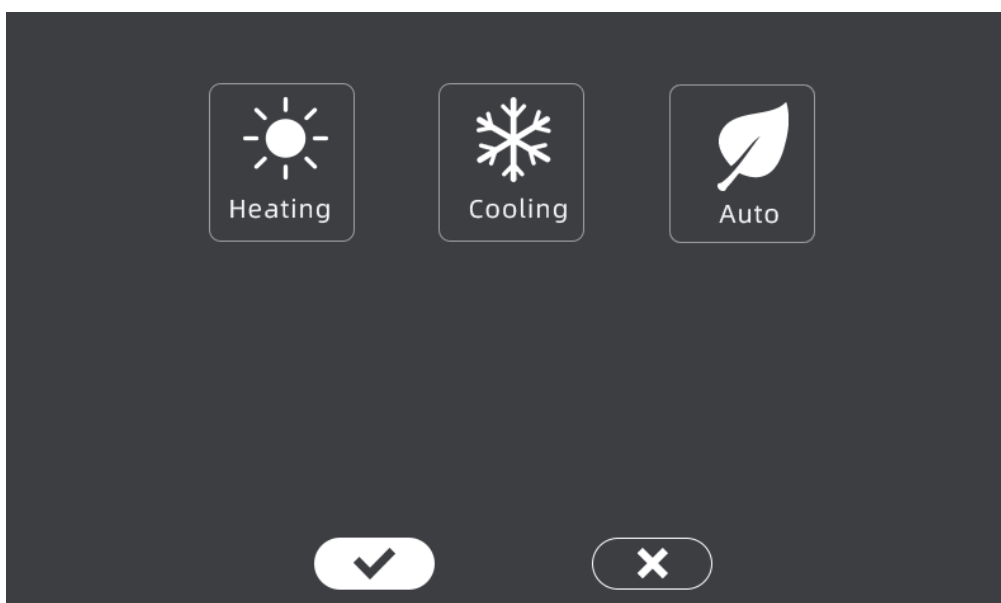


2.1 "ON/OFF" key

- Press the key to turn on or close the unit when screen is on main interface.

2.2 "MODE"key

- Press the key to change mode for AUTO , Cooling , Heating .



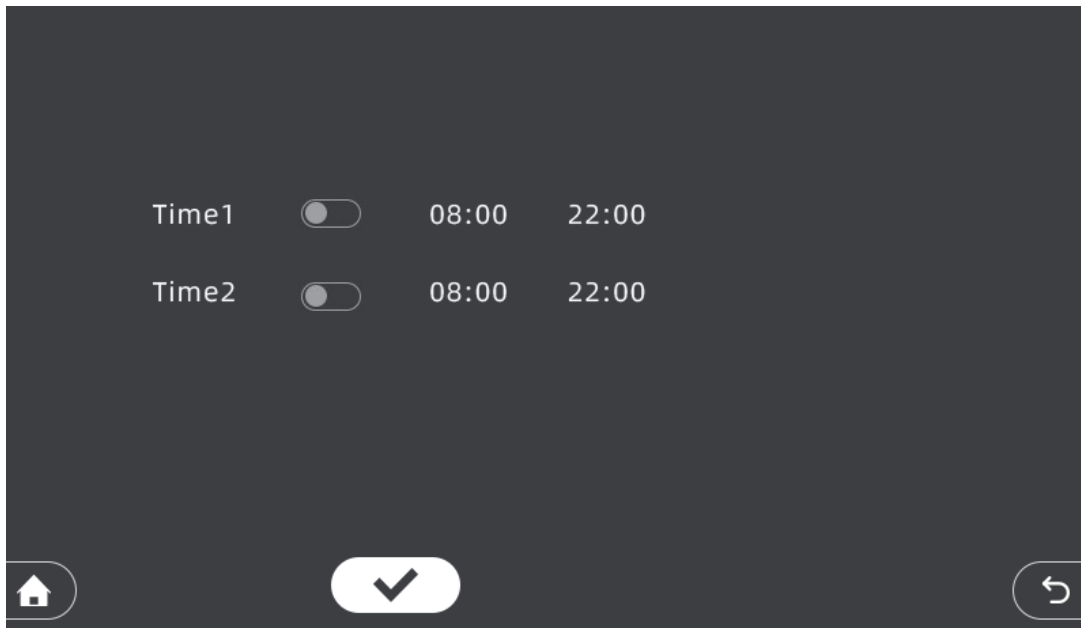
2.3 & Up and down key

- In the main interface, press " " or " " to change the set temperature for the current mode.

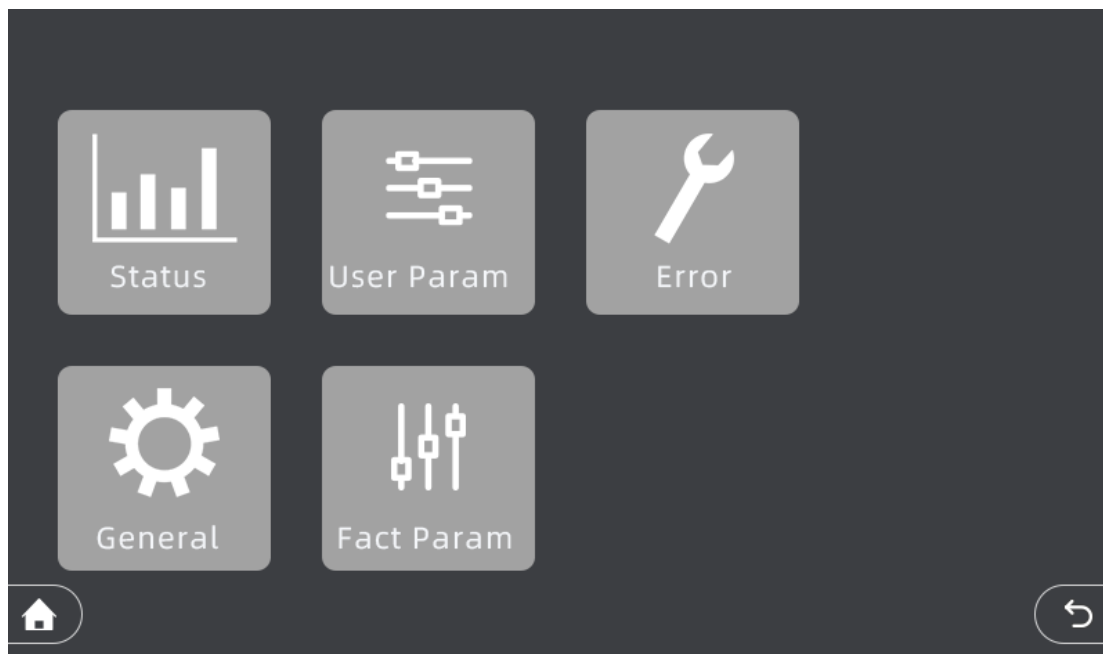


2.4 On/OFF timer setting

There are 2 timer group settings.



2.5 Parameter setting

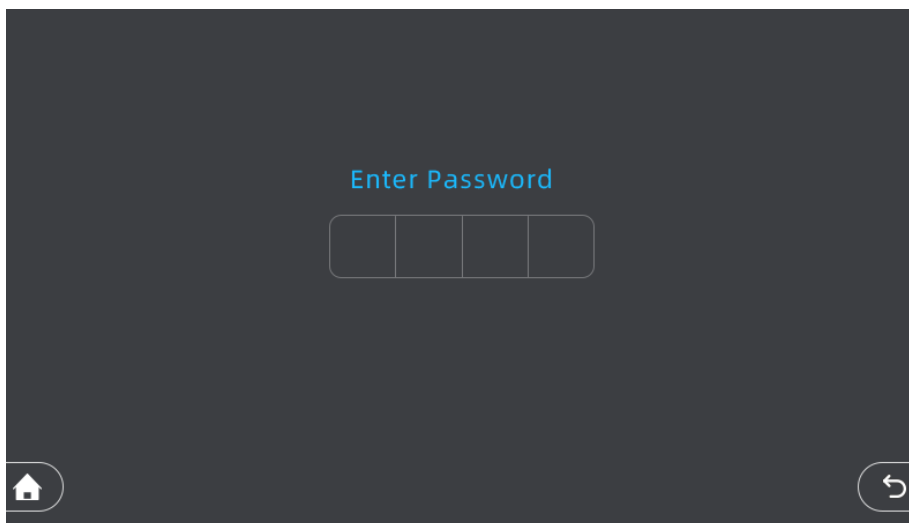




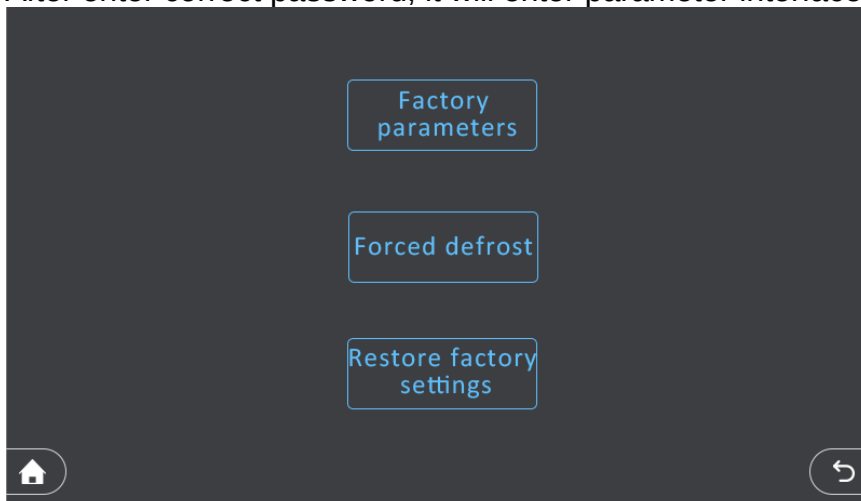
Fact Param

Factory parameter setting.

It needs password to enter setting interface. Password 4180.



After enter correct password, it will enter parameter interface.

**2.6****WIFI connection**

In main interface, press  icon, with a buzzer sound "Beep", it can enter Wi-Fi configuration.

3 Wi-Fi functions

3.1 APP download and installation

- Enter the App Store to search Smart Life APP, then download and install it.



- You also can scan following QR code, download and install Smart Life APP through Android mobile phone or iPhone. But please attention to select “Ordinary Download” when use Android mobile phone.



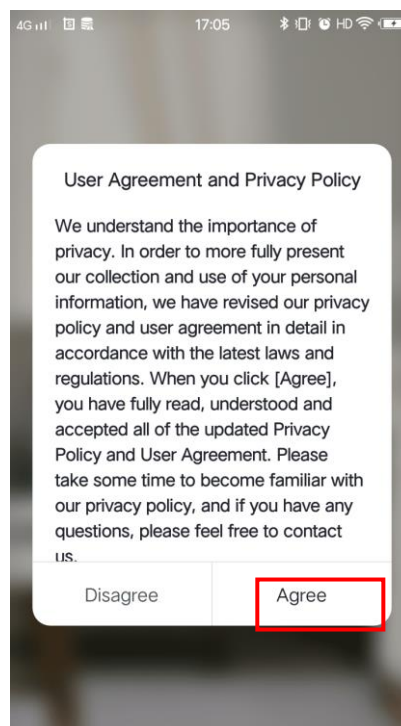
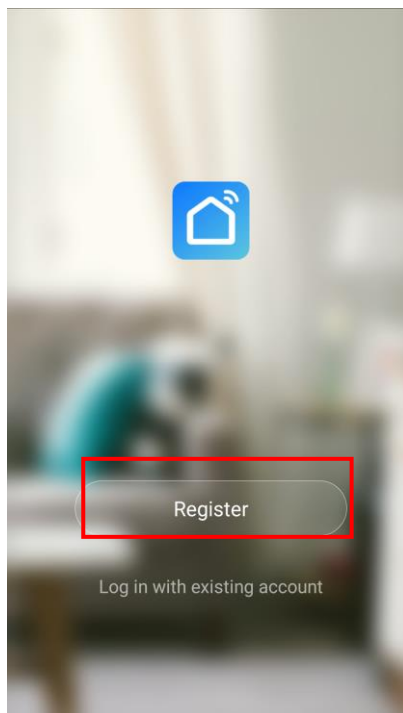
3.2 Start APP

After installation, click the desktop icon "Smart Life" to launch the software.



3.3 Users registration

- For the first time to use the "Smart Life" software, users need to register. Click "Register" to enter the registration interface, and Register with the mobile phone number or email according to the system prompts.



No SIM 15:55

Register

China

Mobile Number/Email

Get Verification Code

☒ I Agree [User Agreement](#) and [Privacy Policy](#)




No SIM 15:54

Enter Verification Code

Verification code has been sent to your mobile phone: 86-13557563914 Resend(42s)

3.4 Create family model

- After successful registration, the software will jump to the "create family" interface.
- Operating as follows : click "create family" → setting "family name" → setting "family location" → add room → click "completed"

15:46

Turn on your smart life

Create family

Log out

15:47

Add family

Completed

Family name jzw

Family location 南海区桂城街道

Smart devices in rooms:

Living Room

Bedroom

Second Bedroom

Dining Room

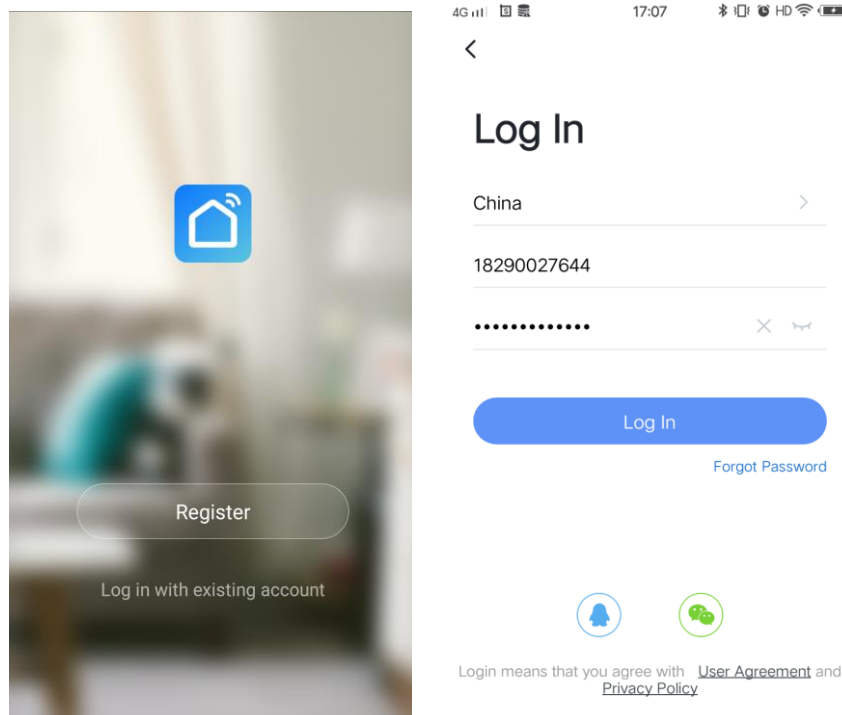
Kitchen

Study Room

Add another room

Afterwards, you can change the room settings at any time

- If you have already registered, simply click "Log om with existing account" to log in.



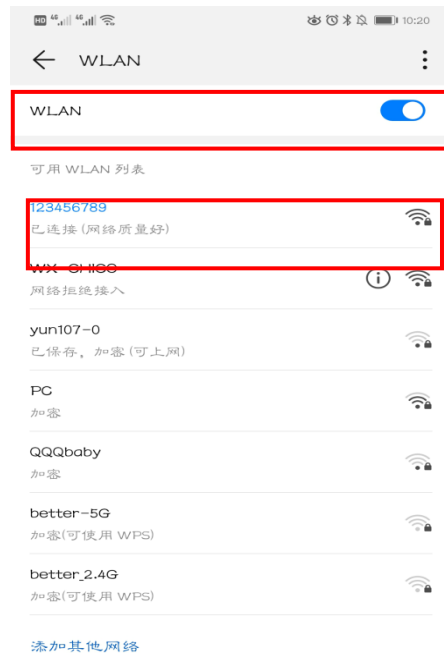
3.5 Device connection process

- **Smart configuration mode**

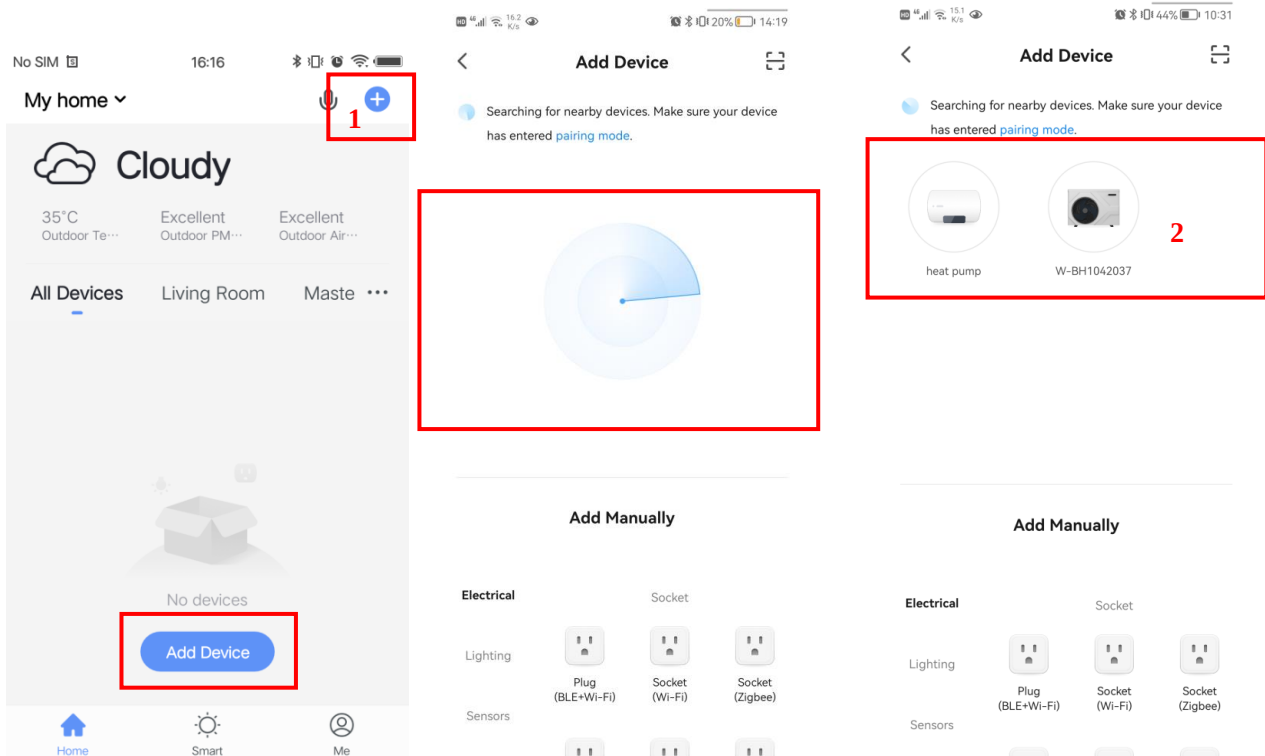
- ◆ Firstly, in main interface, press  icon, with a buzzer sound "Beep" and the icon"  " flashing fast, it can enter Wi-Fi configuration.

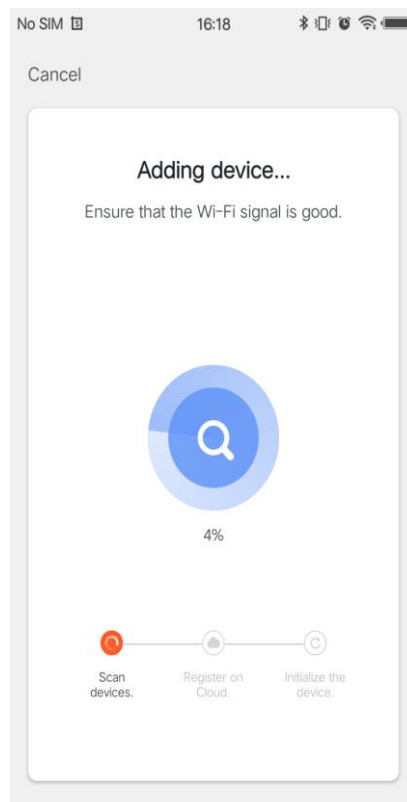


- ◆ Secondary, Turn on the Wi-Fi function of the phone and connect to the Wi-Fi hotspot. The Wi-Fi hotspot must be able to connect to the Internet normally.

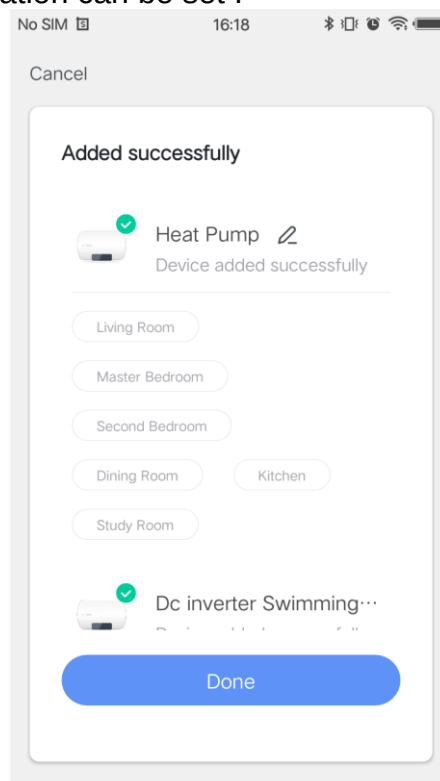


- ◆ Thirdly, open the APP "Smart Life" into the main interface, and click “+” at the top-right corner to enter the device searching interface. Your device will appear and click it.



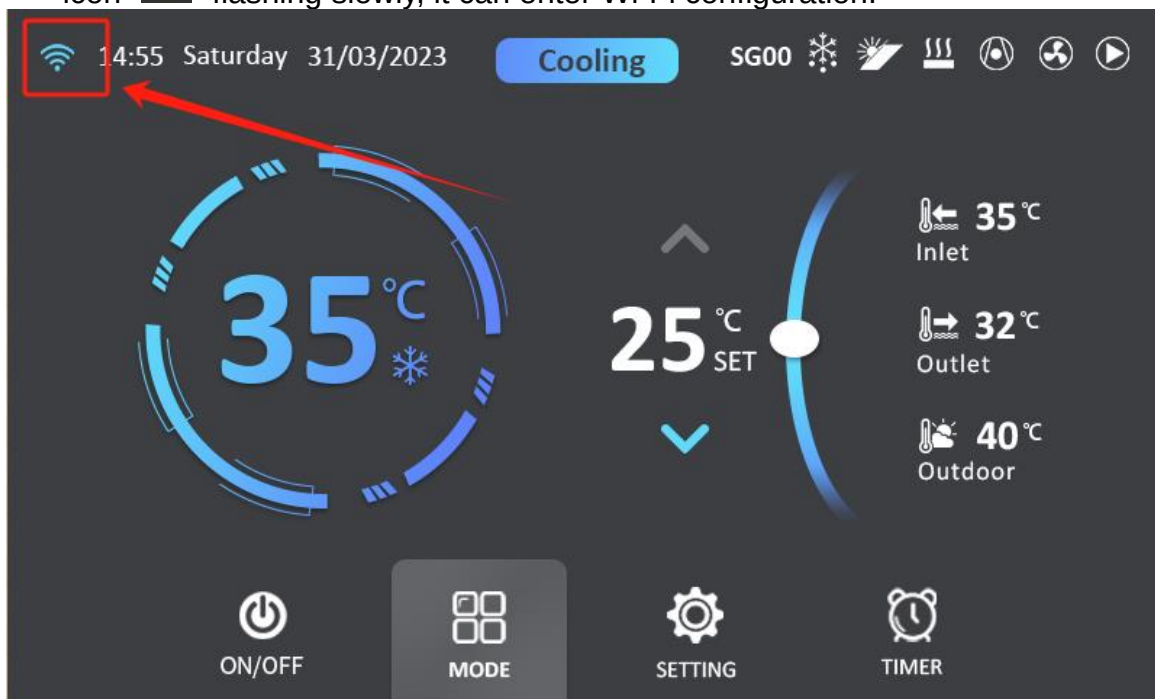


- ◆ Fourthly, when "Scan device", "Register on cloud" and " Initialize the device " are all completed, the connection is successful, and the system prompts "Device added successfully". In the interface, the device name and the device installation location can be set .

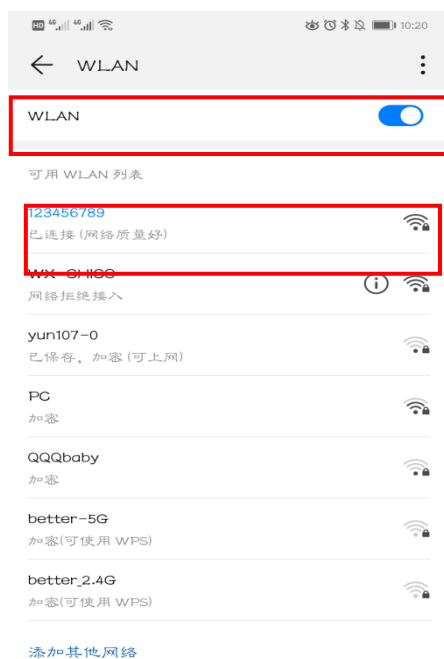


- **Manual configuration mode**

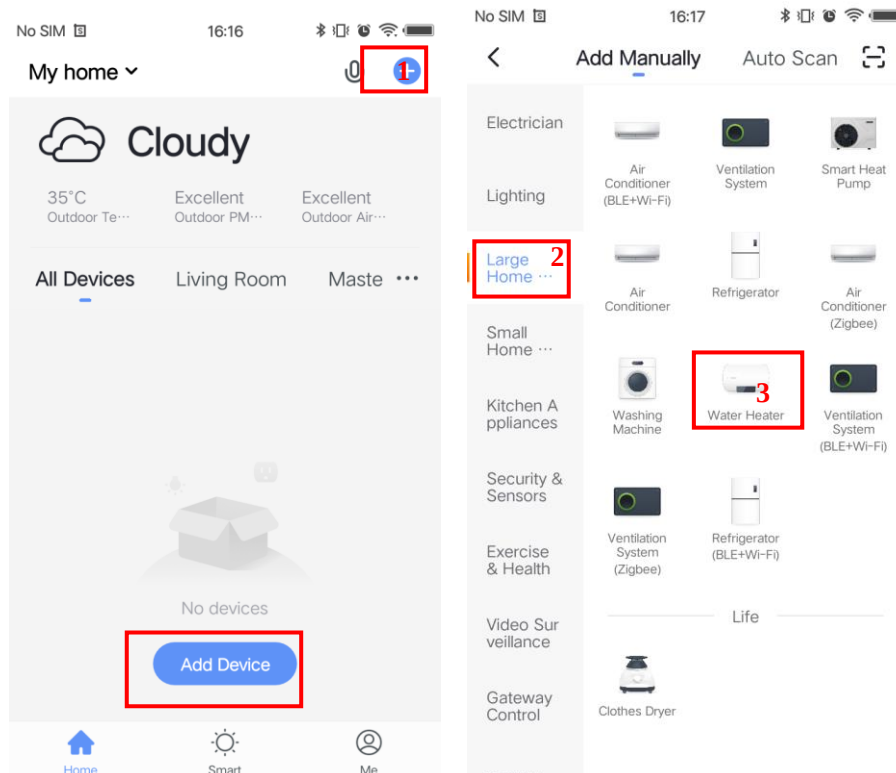
- ◆ Firstly, in main interface, press  icon, with a buzzer sound "Beep" and the icon " flashing slowly, it can enter Wi-Fi configuration.



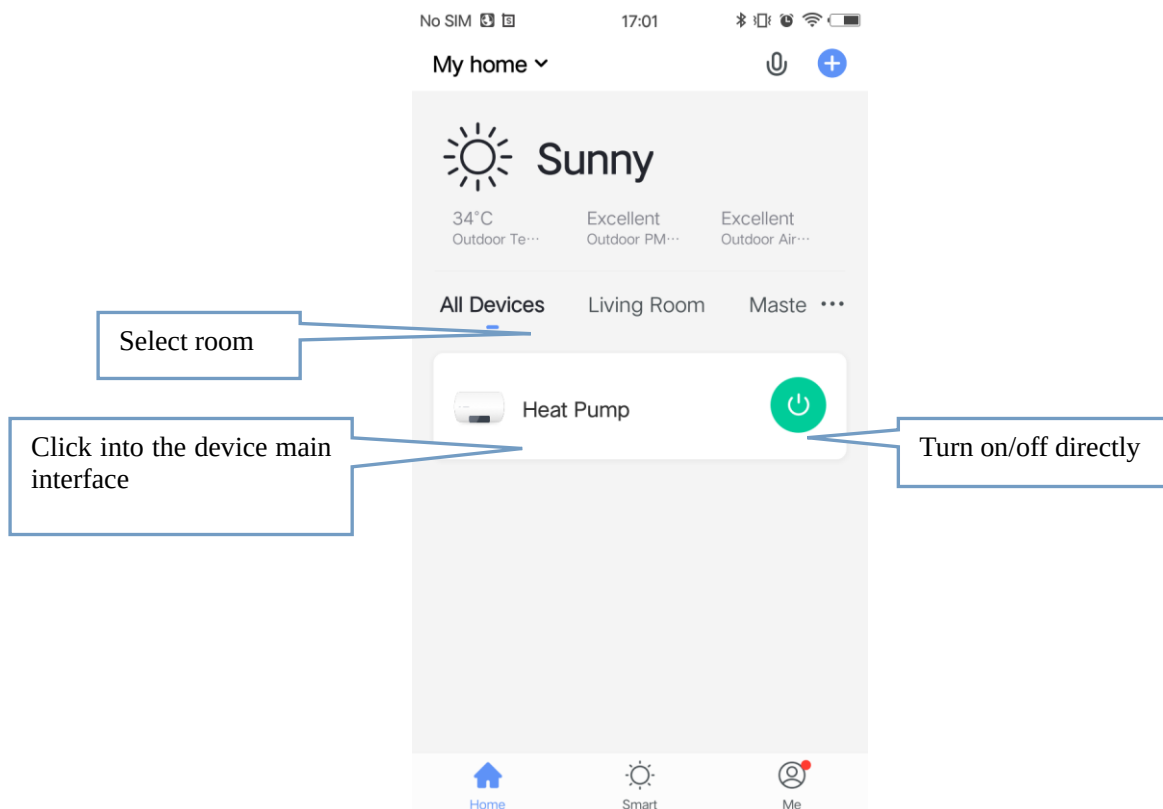
- ◆ Secondary, Turn on the Wi-Fi function of the phone and connect to the Wi-Fi hotspot. The Wi-Fi hotspot must be able to connect to the Internet normally.



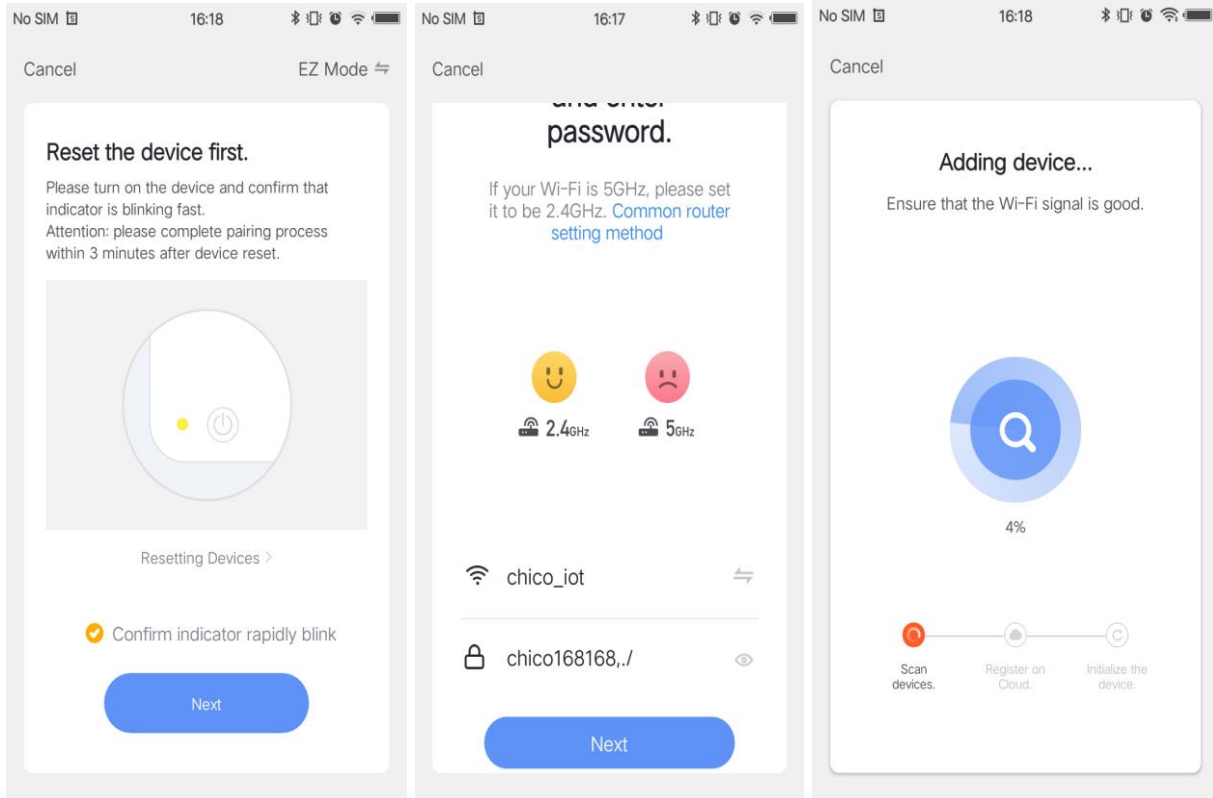
- ◆ Thirdly, open the APP "Smart Life" into the main interface, and click “+” at the top-right corner to enter the devicetype selection interface. Select “Water Heater” from the “Home Appliances” bar. You can also click “Add Device” if no device.



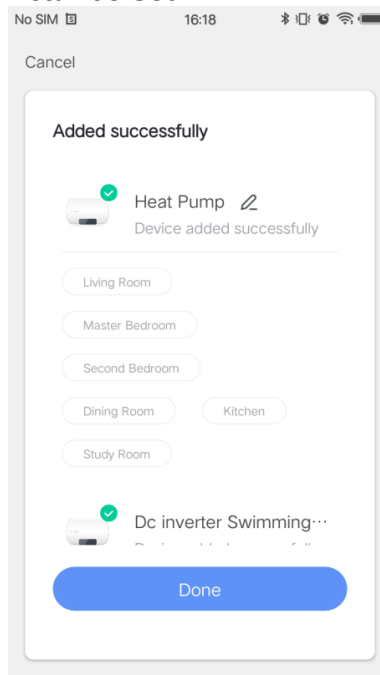
- ◆ If you have already added devices, the APP main interface is shown in the following figure. You can see all the devices that have been added. The drop-down refreshes the connection status.



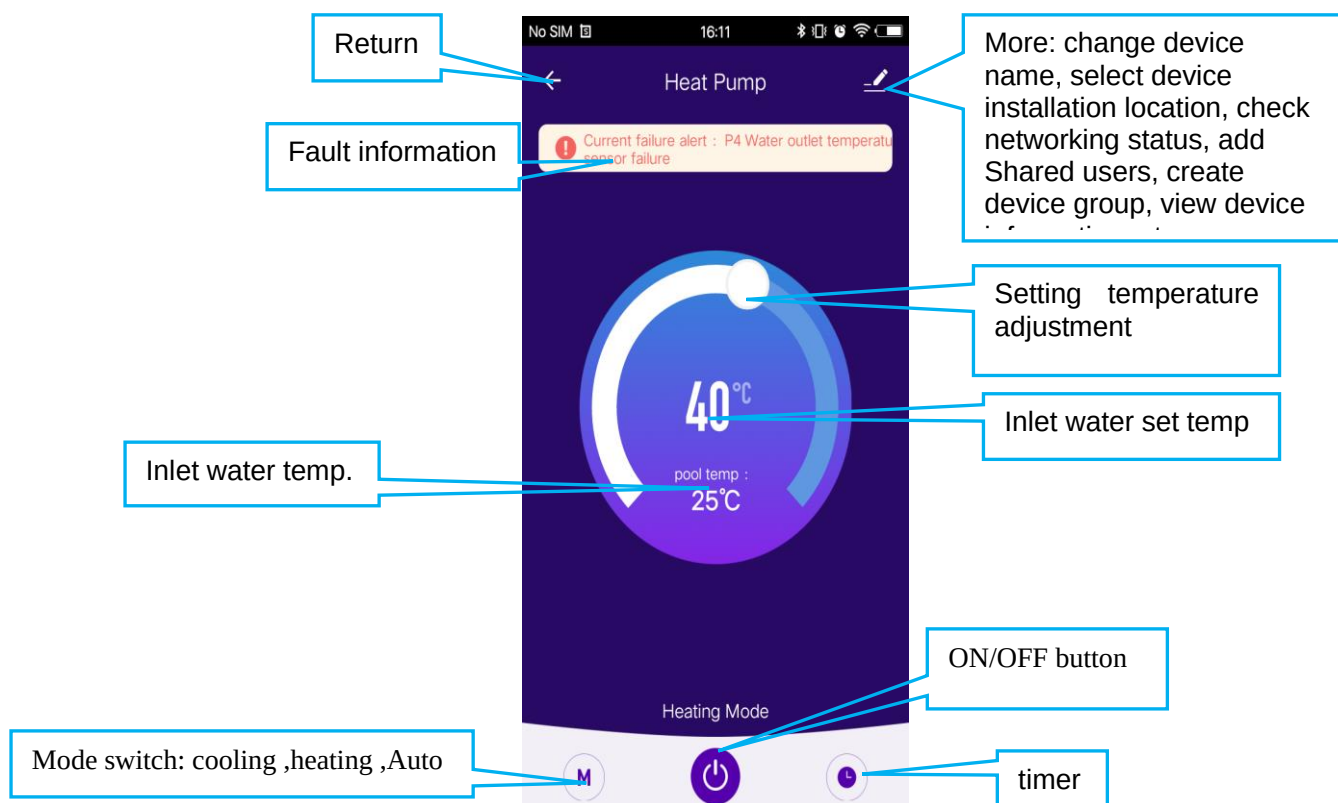
- ◆ Fourthly, after entering the interface of adding device, confirm again that the  icon is flashing, and then Click **"Confirm indicator rapidly blink"**. The Wi-Fi connection interface pops up, enter the Wi-Fi password (it must be consistent with the Wi-Fi connected by the phone), and click **"Next"** to enter the connection state.



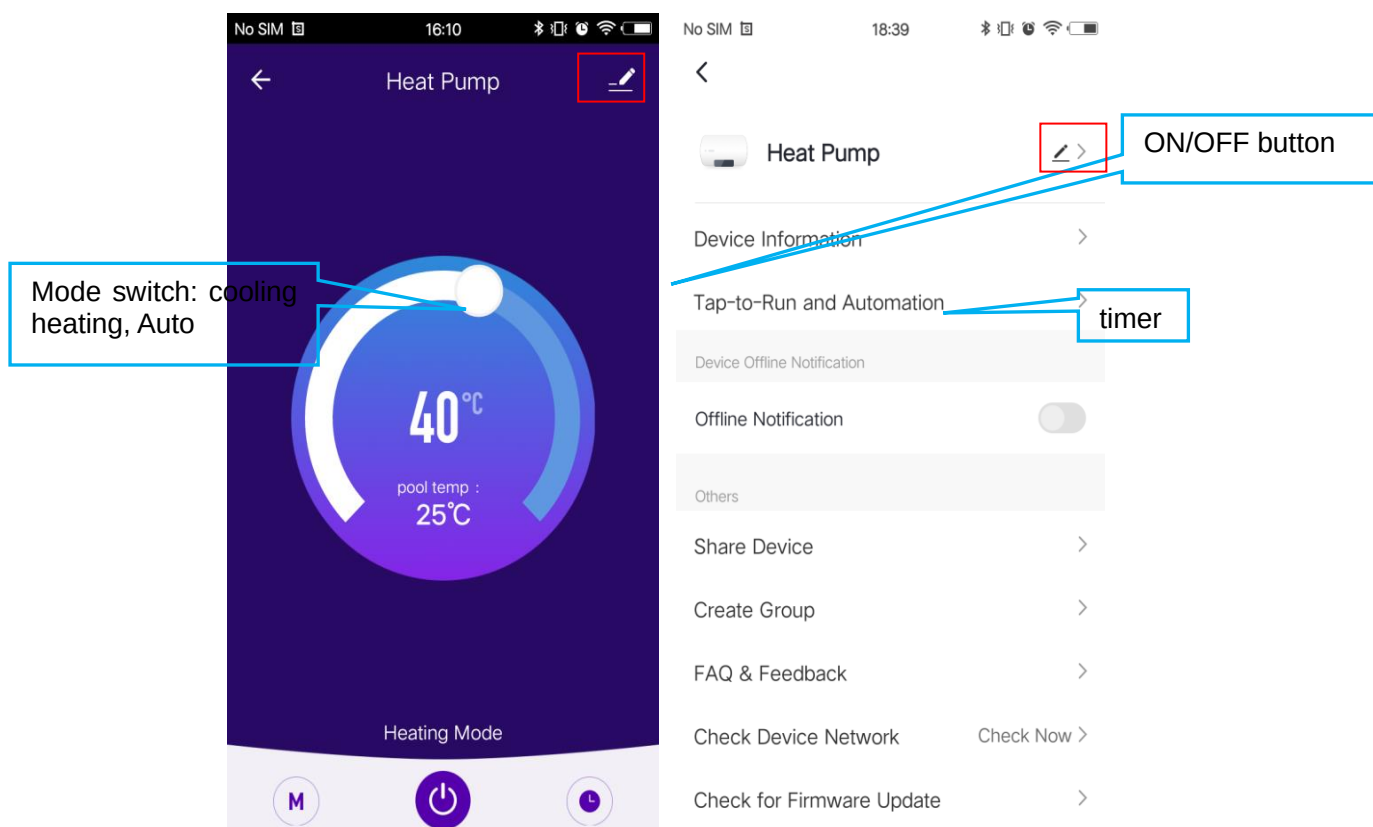
- ◆ Fifth, when "Scan device", "Register on cloud" and " Initialize the device " are all completed, the connection is successful, and the system prompts "Device added successfully". In the interface, the device name and the device installation location can be set .



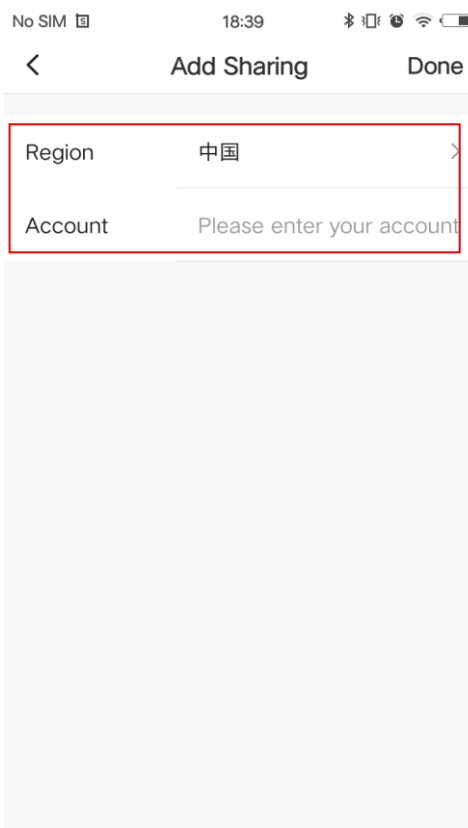
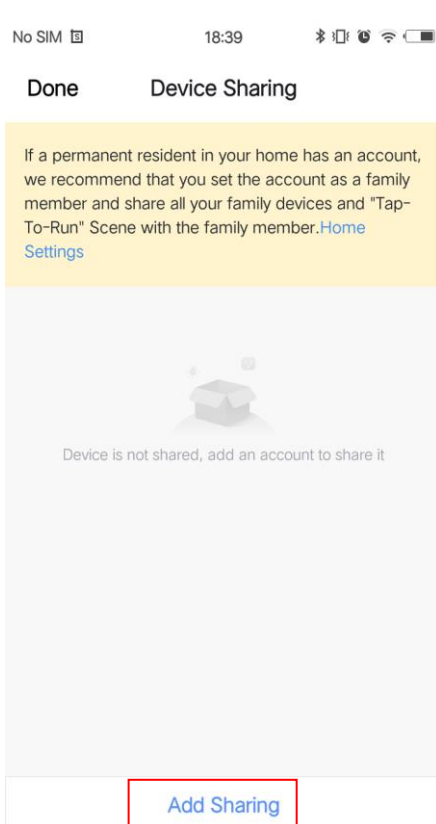
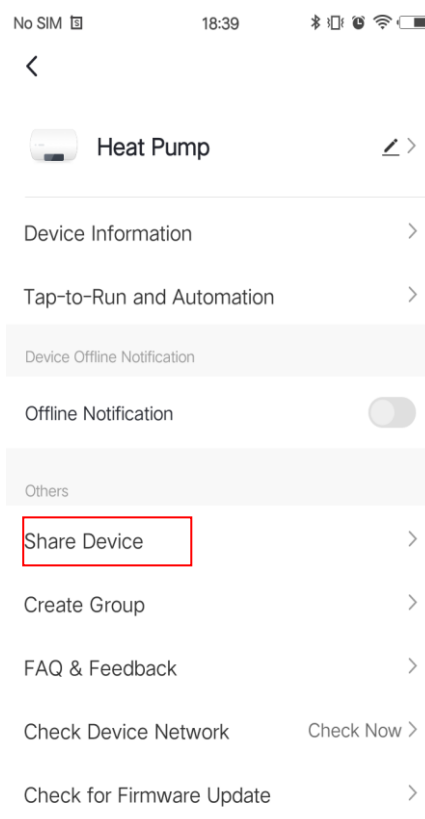
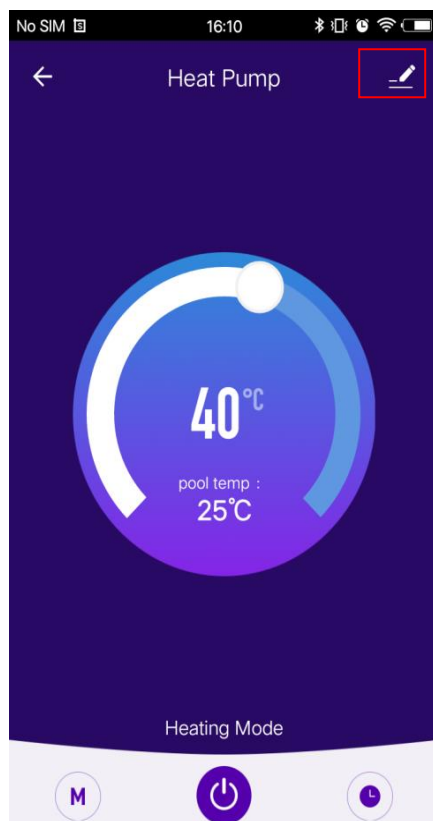
3.6 Ice bath operation interface



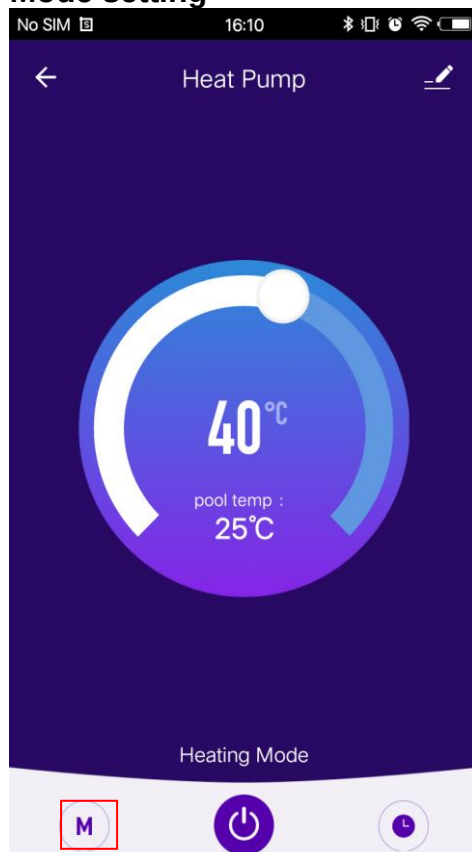
● Modify device name



- Share device



- **Mode setting**



Working Mode

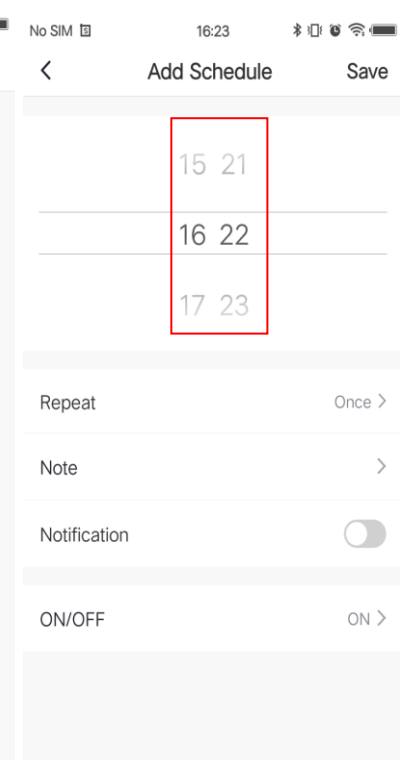
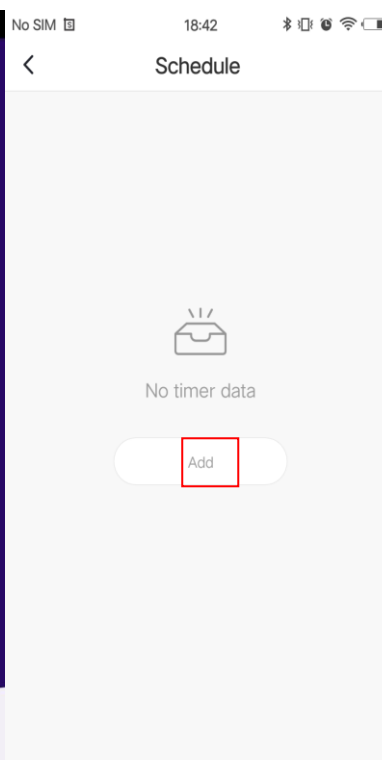
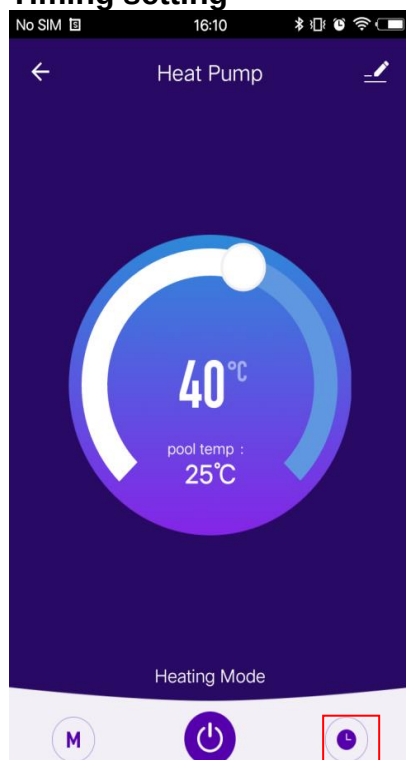
Auto Mode

Cooling Mode

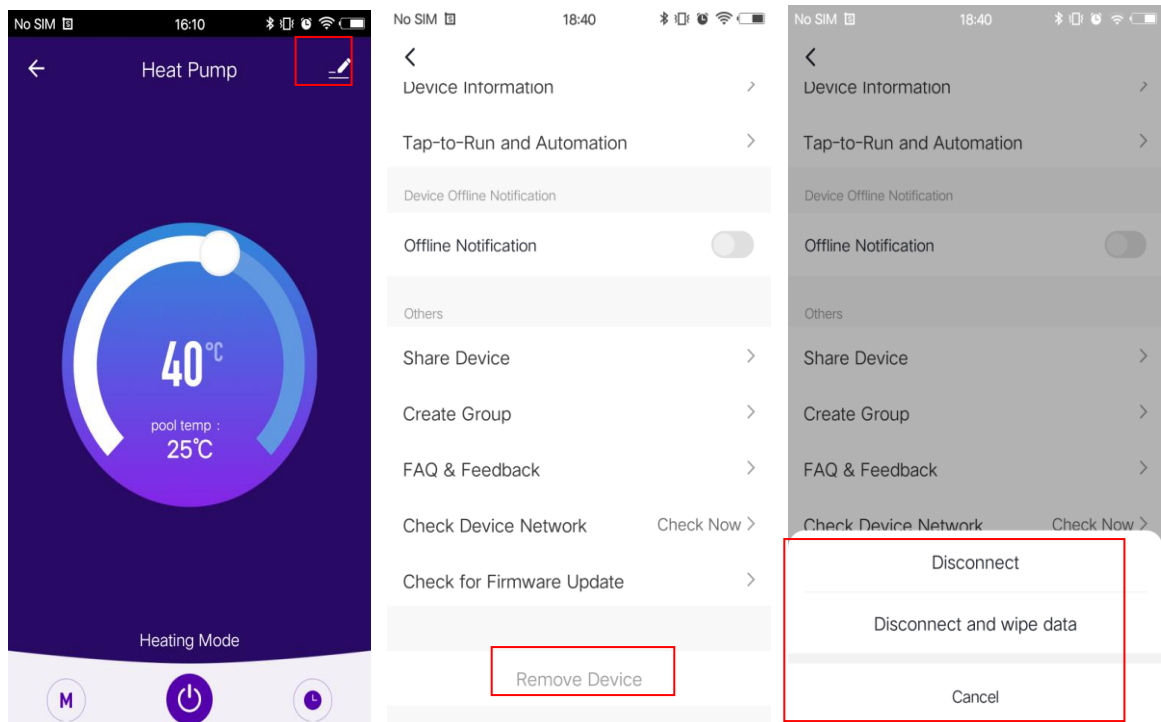
Heating Mode

Done

- **Timing setting**



● Remove device



4 Operation modes

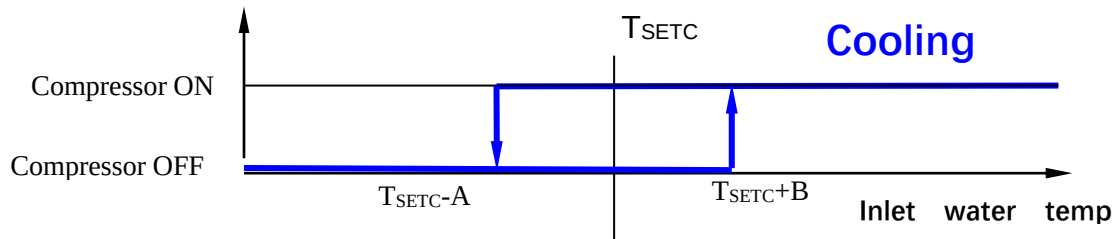
4.1 Cooling mode

- **Temperature setting**

- ◆ Inlet water temperature set value T , range 3-28°C(parameter P01), Initial set temperature is 27°C;

- **Operation control**

- ◆ 4-way valve is powered off, water pump opens, Whether compressor open or shutdown is decided by the value of Inlet water temperature T_{IN} and T_{SETC}



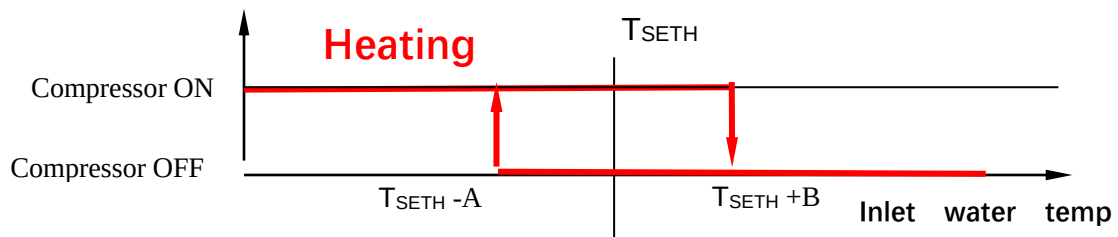
4.2 Heating mode

- **Temperature setting**

- ◆ Inlet water temperature set value T , range 15-45°C(parameter P02), Initial set temperature is 40°C;

- **Operation control**

- ◆ 4-way valve is powered on, water pump opens, Whether compressor open or shutdown is decided by the value of Inlet water temperature T_{IN} and T_{SETH}



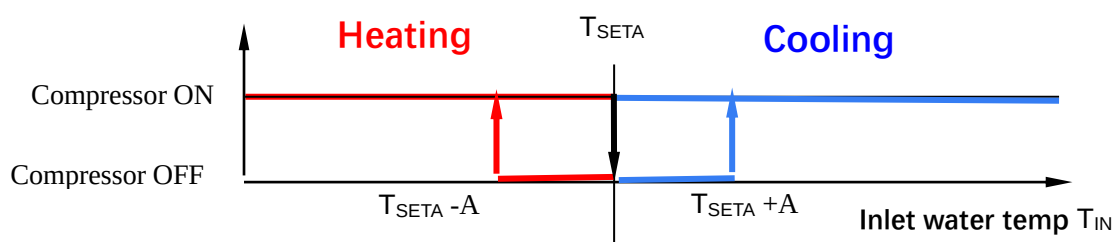
4.3 Auto mode

- **Temperature setting**

- ◆ Inlet water temperature set value T , range 3-45°C(parameter P08), Initial set temperature is 40°C;

- **Operation control**

- ◆ Heating: 4-way valve is powered on, water pump opens, Whether compressor open or shutdown is decided by the value of Inlet water temperature T_{IN} and T_{SETA} .
- ◆ Cooling: 4-way valve is powered off, water pump opens, Whether compressor open or shutdown is decided by the value of Inlet water temperature T_{IN} and T_{SETA}



4.4 Defrosting

- **Condition to enter defrosting**

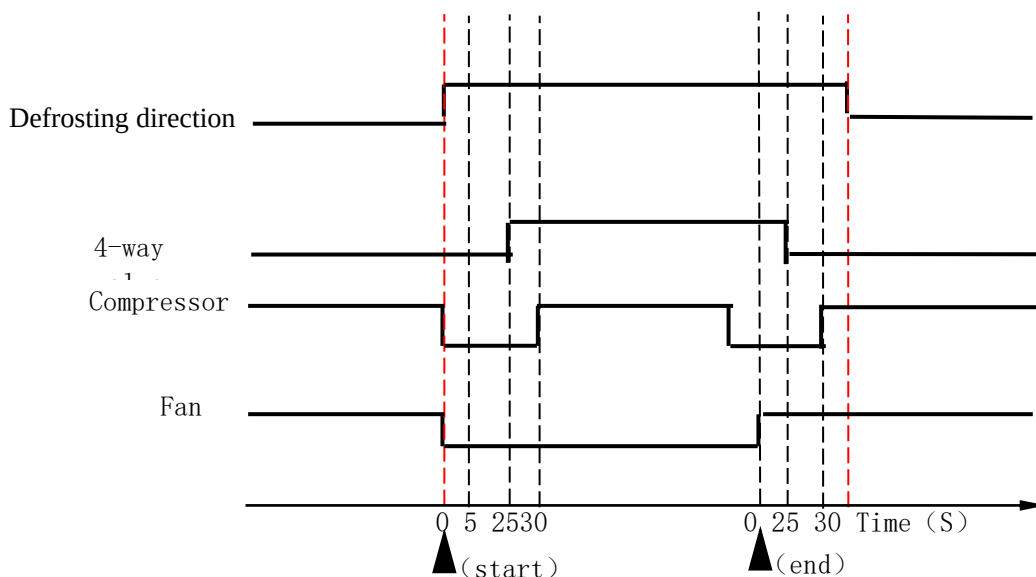
- ◆ In heating mode, units operates for a total of 40 minutes (parameter 03, range: 30-90minutes) under condition coil temperature $<13^{\circ}\text{C}$ (parameter 05,range: 2- 30°C), it enters defrosting when tested coil temperature $\leq -7^{\circ}\text{C}$,(parameter 05,range: $-30-0^{\circ}\text{C}$);
- ◆ When coil temperature failure (PP3) occurs and $T_{\text{ambient}} < 10^{\circ}\text{C}$, defrosting will be changed to regular defrosting for 6 minutes.

- **Condition to quit from defrosting**

- ◆ When coil temperature. $>13^{\circ}\text{C}$ (parameter 05 ,range: $2^{\circ}\text{C}\sim 30^{\circ}\text{C}$), or defrosts for 6 minutes (parameter 06, range:1~12 minutes), it quits from defrosting.

- **Defrosting action**

- ◆ Following action begins when it meets defrosting condition:
 - ✓ Compressor and outdoor fan stop running. Send the defrosting starting signal to the remote controller
 - ✓ At 25 seconds, the 4-way exchange valve will be powered off;
 - ✓ At 30 seconds, compressor starts;
 - ✓ Water pump operates normally.
- ◆ Following action begins when it meets defrosting quitting condition:
 - ✓ When system meets the defrosting quitting condition, it quits from defrosting. Compressor stops running and outdoor fan begins to run. 4-way exchange valve will be powered on 25 seconds later.
 - ✓ Compressor starts and recovers to normal heating after fan runs for 30 seconds. The heating operation time is reset. Send the defrosting closing signal to remote controller and stops issuing defrosting signal.



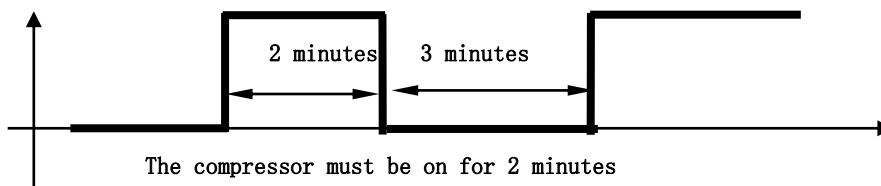
- **Abnormal completion of defrosting**

- ◆ If unit stops during defrosting, it will keep defrosting until defrosting ends.
- ◆ Low pressure protection is shielded during defrosting and it quits from defrosting to normal heating, low pressure switch is detected for 1 minute later.

5 Relay output control

5.1 Compressor

- Safe time of compressor start to stop and start to start (except for defrosting procedure);
- After the same compressor stops, wait 3 minutes to restart.
- There must be a time span of 5 minutes between two starts for the same compressor.



5.2 4-way valve

- It is powered on in cooling mode. When the unit starts, the 4-way valve will turn on with water pump. When system requires unit to turn off, it will turn off in 2 minutes after the compressor stops.
- It delays 2 minutes for the 4-way exchange valve to stop when it changes from cooling mode to heating mode;
- The compressor starts only after the 4-way exchange valve starts for 5 seconds when it changes from heating mode to cooling mode;
- It delays 2 minutes for the 4-way exchange valve to stop when it changes from cooling mode to shutdown.

5.3 Fan

- When the unit starts, the fan will turn on 3 seconds earlier than compressor.
- When system requires unit to turn off, the fan will stop 30s later than compressor.
- When defrosting, it is controlled according to the requirement of defrosting.

5.4 Water pump

- When system requires to start, water pump starts before the compressor for 90 seconds.
- When system requires to stop, water pump stops after the compressor and fan stop for 30 seconds.
- During defrosting, water pump dose not stop running.
- **For normal models (parameter10=0), water pump dose not stop at constant temperature. For special models (parameter 10=1), The water pump stops after the compressor is shut off for 30 seconds. Then stop for 15 minutes (parameter 11) and open for 2 minutes n cycle and cycle.**

Note: In the shutdown state, if the pump stops for 12 hours, it needs to be turned on and run for 2 minutes.

LED light:

Machine Turn on, LED on. Turn off, LED off.

5.5 EEV

● Operation

- ◆ Initial opening steps: 350P.
- ◆ EEV resets first when unit turn on. Then EEV is turned to 550P at the beginning and then is turned to initial opening aperture of 350P.

● Normal control logic

◆ In heating mode

EEV action step number will be decided by calculation.

$T_{\text{discharge gas}} - T_{\text{inlet water}} = 35$ (target superheat).

If the target superheat is not 35, then adjust EEV according to table below:

◇ Discharge gas temperature TE	◇ EEV adjustment operation
$TE \geq 120^{\circ}\text{C}(\text{P9})$, and last for 10s	Machine off and Discharge gas overheat protection
$TE - T_{\text{inlet water}} > 40^{\circ}\text{C} (\text{P13}+5^{\circ}\text{C})$	Open 2P each cycle
$(\text{P13}-5^{\circ}\text{C}) 30^{\circ}\text{C} \leq TE - T_{\text{inlet water}} \leq 40^{\circ}\text{C} (\text{P13}+5^{\circ}\text{C})$	Keep it as it is
$TE - T_{\text{inlet water}} < 30^{\circ}\text{C} (\text{P13}-5^{\circ}\text{C})$	Close 2P each cycle

◆ In cooling mode

EEV action step number will be decided by table below:

$T_{\text{inlet water}} \backslash T_{\text{Ambient}}$	$T_{\text{Ambient}} \leq 30$	$30 < T_{\text{Ambient}} \leq 45$	$45 < T_{\text{Ambient}}$
$T_{\text{inlet water}} \leq 10$	y0 (210)	y1 (240)	y2 (280)
$10 < T_{\text{inlet water}} \leq 20$	y3 (250)	y4 (310)	y5 (340)
$T_{\text{inlet water}} > 20$	y6 (300)	y7 (360)	y8 (390)

Discharge temperature protection

◇ Discharge gas temperature TE	◇ EEV adjustment operation
$TE \geq 120^{\circ}\text{C}(\text{P9})$, and last for 10s	Machine off and Discharge gas overheat protection
$TE \geq 112^{\circ}\text{C}(\text{P14}+8^{\circ}\text{C})$	Open 10P each cycle
$TE \geq 110^{\circ}\text{C}(\text{P14}+6^{\circ}\text{C})$	Open 8P each cycle
$TE \geq 108^{\circ}\text{C}(\text{P14}+4^{\circ}\text{C})$	Open 6P each cycle
$TE \geq 106^{\circ}\text{C}(\text{P14}+2^{\circ}\text{C})$	Open 4P each cycle
$TE \geq 94^{\circ}\text{C} (\text{P14})$	Keep it as it is
$TE < 94^{\circ}\text{C} (\text{P14}-5^{\circ}\text{C})$	Return normal control logic

● EEV action period:

EEV acts once every 30 seconds.

- During refrigeration, defrosting and compulsory defrosting, EEV fixed opening aperture is 400P(parameter P24).

6 System protection

6.1 Water flow failure

- It begins to detect after water pump starts for 30 seconds. If water flow switch disconnection is detected for continuous 5 seconds, it is determined to be a water flow failure. Compressor and fan stop while water pump dose not stop.
- When water flow failure occurs, if water flow switch closing is detected, unit recovers to normal operation.
- When failure occurs, failure code “FFF” is displayed on the wiring panel.

6.2 Anti-freeze protection in winter

- In standby status, detects inlet water temp and ambient temp.
- When ambient temp $\leq 0^{\circ}\text{C}$ and inlet water temp $\leq 15^{\circ}\text{C}$, it enters to the one class anti-freeze status. Unit will automatically open the water pump mode.
- If ambient temp. $\leq 0^{\circ}\text{C}$ and inlet water temp $\leq 2^{\circ}\text{C}$, it enters to the two class anti-freeze status. Unit will automatically open to operate the heating mode.
- When inlet water temp $\geq 15^{\circ}\text{C}$ or ambient temp $> 2^{\circ}\text{C}$, it quits from anti-freeze protection.
- If ambient temp failure occurs, whether it enters anti-freeze protection is decided by inlet water temp. If inlet water temp failure occurs, whether it enters anti-freeze protection is decided by ambient temp. (Inlet water temp failure occurs , It enters the one class anti-freeze status only)
- If both inlet water temp and ambient temp failure occurs, anti-freeze protection will not be treated.
- While in anti-freeze protection, failure code “AFP” is displayed on the wiring panel.

6.3 Inlet water temperature sensor failure

- When inlet water temperature sensor is detected to be short-circuited or disconnected at any time, it is determined to be a inlet water temperature sensor failure then unit stops.
- When it is detected that inlet water temperature sensor recovers to normal, unit will recover to normal operation.
- When failure occurs, failure code “PP1” is displayed on the wiring panel.

6.4 Discharge temperature sensor failure

- After unit is powered on, if it is detected at any time that discharge temperature sensor short-circuiting occurs or discharge temperature sensor disconnects after compressor starts and works for 1 minute, it will be determined to a discharge temperature failure and unit will stop for protection.
- This failure can be recovered automatically.
- When failure occurs, failure code “PP2” is displayed on the wiring panel.

6.5 Coil temperature sensor failure

- When coil temperature sensor is detected to be short-circuited or disconnected at any time, it is determined to be a coil temperature sensor failure then and unit works normally. Defrosting control will be changed to timer defrosting .
- When it is detected that coil temperature sensor recovers to normal, unit will recover to normal operation.
- When failure occurs, failure code “PP3” is displayed on the wiring panel.

6.6 Ambient temperature sensor failure

- When outdoor temperature sensor is detected to be short-circuited or disconnected at any time, it is determined to be a ambient temperature sensor failure then and unit works normally. Anti-freeze protection is decided by inlet water temperature.
- When it is detected that outdoor temperature sensor recovers to normal, unit will recover to normal operation.
- When failure occurs, failure code “PP5” is displayed on the display panel .

6.7 Low pressure protection

- After compressor starts for 5 minutes, if system low pressure switch disconnection is detected for continuous 10 seconds, it is determined to be a system low pressure failure. Then unit stops.
- When low pressure switch closing is detected, unit will quit from failure protection and recover to normal operation (without 3 times of system failure in 30 minutes).
- If system failure occurs for continuous 3 times within 30 minutes, it will not recover unless power off.
- When failure occurs, failure code “EE2” is displayed on the wiring panel.

6.8 Excessive discharge temperature protection

- Detect excessive discharge temperature protection after compressor starts for 1 minute, if it is detected that discharge temperature $\geq 120^{\circ}\text{C}$ (parameter 9) for continuous 5 seconds, the unit will stop for protection.
- When it is detected that discharge temperature $\leq 90^{\circ}\text{C}$, it will quit from this protection.
- If this failure occurs for 3 times within 30 minutes, it will not be recovered unless it is powered off (the former 2 times can be recovered automatically.).
- When failure occurs, failure code “EE5” is displayed on the display panel.

6.9 Communication failure

- Within 20 seconds when power on for the first time, if main board dose not receive any communication signal from display panel, it is determined to be disconnected with remote control. System controls the operation just according to emergency switch signal
- In normal states ,if display panel dose not receive any signal from main board for continuous 10 minutes, it is determined to be a communication failure and failure code “EE6” is displayed.

7 Other functions

7.1 Power-off memory

- System will record the parameters all the way.
- System can keep ON/OFF status and electric heating status in memory.
- After abnormal power-off or unit closure, once it is powered on, the system will be in standby status or remaining status before power-off.

7.2 Error code table

Protection/Failure	Failure code	Indicator lamp for unit operation /failure
Standby		Off
Normally operation		light
Inlet water temperature sensor failure	PP1	
Discharge temperature sensor failure	PP2	
Coil temperature sensor failure	PP3	
Ambient temperature sensor failure	PP5	
One class anti-freeze	AFP	
Two class anti-freeze	AFP	
Water flow failure	FFF	
Low pressure protection	EE2	
Excessive discharge temperature protection	EE5	
Communication failure	EE6	
Defrosting direction	 flashing	

7.3 System parameter

Code	Meaning	Range	Default	Note
P01	Set temperature for cooling	3~45°C (50°F~113°F)	27(81°F)	adjustable
P02	Set temperature for heating	3~45°C (50°F~113°F)	27(81°F)	adjustable
P03	Defrosting cycle	30-90 min	40 min	adjustable
P04	Temperature setting for starting defrosting	-30-0°C	-7°C	adjustable
P05	Temperature setting for exiting defrosting	2-30°C	20°C	adjustable
P06	defrosting running time setting	1-12 min	8 min	adjustable
P07	Model options	0(cooling)\1 (C&H) \2 (heating)	1	adjustable
P08	Set temperature for AUTO	3~45°C (50°F~113°F)	27(81°F)	adjustable
P09	Discharge temperature setting for protection	85~120°C (185°F~248°F)	120(248°F)	adjustable
P10	water pump running mode	0 (normal) \1 (special)	1	adjustable
P11	Temperature difference for restart	1~4°C(2°F~7°F)	1(2°F)	adjustable
P12	The max. water set temperature	30~50°C (86°F~122°F)	40 (104°F)	adjustable
P13	Target temp. for (T discharge gas — T inlet water)	5~100	35(63°F)	adjustable
P14	Discharge temp. for adjusting in cooling	80~120	98 (208°F)	adjustable
	Machine selection	110/220	110	adjustable

				adjustable
P15	EEV opening in cooling (y0)(110V)	100~500	210	adjustable
P16	EEV opening in cooling (y1)(110V)	100~500	240	adjustable
P17	EEV opening in cooling (y2)(110V)	100~500	280	adjustable
P18	EEV opening in cooling (y3)(110V)	100~500	250	adjustable
P19	EEV opening in cooling (y4)(110V)	100~500	310	adjustable
P20	EEV opening in cooling (y5)(110V)	100~500	340	adjustable
P21	EEV opening in cooling (y6)(110V)	100~500	300	adjustable
P22	EEV opening in cooling (y7)(110V)	100~500	360	adjustable
P23	EEV opening in cooling (y8)(110V)	100~500	390	adjustable
P15	EEV opening in cooling (y0)(220V)	100~500	260	adjustable
P16	EEV opening in cooling (y1)(220V)	100~500	290	adjustable
P17	EEV opening in cooling (y2)(220V)	100~500	330	adjustable
P18	EEV opening in cooling (y3)(220V)	100~500	300	adjustable
P19	EEV opening in cooling (y4)(220V)	100~500	360	adjustable
P20	EEV opening in cooling (y5)(220V)	100~500	390	adjustable
P21	EEV opening in cooling (y6)(220V)	100~500	350	adjustable
P22	EEV opening in cooling (y7)(220V)	100~500	410	adjustable
P23	EEV opening in cooling (y8)(220V)	100~500	440	adjustable
P24	EEV opening for defrost	100~500	400	adjustable
P25	Mode selecting for EEV opening in cooling	0 (110V)~ 1(220V)	0	adjustable

7.4 System state

A10	Inlet water temperature	-9℃~99℃		measured value
A02	Discharge temperature	-9℃~128℃		measured value
A03	Coil temperature	-9℃~99℃		measured value
A04	Ambient temperature	-9℃~99℃		measured value
A05	EEV steps	0~500		measured value